Product 2: Vendor-Agnostic PV Inverter Controller — Research Report

Prepared for OnyxLeaf (Rawalpindi/Islamabad) — 30 October 2026. Research performed by web search and direct fetches of the cited pages. Every claim carries a link to the page opened; where a number or fact could not be confirmed from a fetched page it is explicitly marked **not verified**.

Scope

This report covers the edge-layer product only: a controller that reads, normalizes, and (where safely possible) controls any brand of PV inverter and BESS, exposing one uniform interface upward. The meeting notes define the product as: each inverter brand exposes current, voltage, active and reactive power over serial with "its own pattern" (Sungrow one pattern, Growatt another); the product should "detect all patterns and adapt accordingly", counter vendor lock-in ("buy inverter of my company"), fix the OEM supplier's wrong battery data, and be either "a controller (reading AND control for any brand's hardware) OR a frontend+backend", with the compatibility claim "works with all systems" and BESS included.

Our reading: the controller and the frontend/backend are not alternatives — they are two halves of one product. The **edge controller** is a small box that sits on the inverter's RS485/CAN/USB ports, speaks each brand's dialect, and normalizes everything into one data model; the **backend + frontend** is the cloud that aggregates fleets of these boxes and gives installers/owners a single pane of glass across mixed-brand sites. The interface exposed upward to Product 1 (the TOU-aware EMS/optimizer, out of scope here) is: **normalized telemetry** (PV power, grid import/export, load, battery V/I/SoC/temperature, per-phase AC quantities, alarms) plus **writable normalized setpoints** (charge/discharge power or current limits, time-of-use windows, work mode, export limit), with the controller translating each setpoint into the brand-specific register writes or serial commands and enforcing fail-safes. Optimization logic, tariff awareness, and scheduling are explicitly out of scope — this product's job is to make any inverter look like the same inverter.

Refined problem statement

Pakistani solar sites increasingly mix inverter brands (a Growatt string inverter beside an Inverex/Voltronic hybrid beside a Deye-based unit), and each brand ships its own protocol, register layout, Wi-Fi dongle, and cloud app, so an owner or installer with N brands needs N monitoring apps and has no single point of control. Vendors exploit this: the sanctioned path to monitoring and control is their own dongle and cloud, which locks the customer into the vendor's batteries and accessories, and the dongle physically occupies the one RS485/USB port a third-party controller would need. OEM-rebadged LFP battery packs frequently report wrong state-of-charge to the inverter and to the user, so decisions (and warranties) rest on bad data. The falsifiable claims behind the product: (1) ≥80% of the inverters installed in Pakistan speak one of fewer than ~10 protocol families, almost all already reverse-engineered in open source; (2) a curated driver library plus automatic device identification can connect to an unknown site in under 15 minutes without a laptop-wielding protocol engineer; (3) enough of the common brands expose writable control registers that a cross-brand control product (not just monitoring) is feasible. Each of these is testable on a bench and in a 5-site pilot, and claim (3) is the kill test (see §Pricing, risks and kill criteria).

The protocol problem

The point of this section: there is no single "serial pattern" to detect — there are roughly three protocol families (standard Modbus with per-vendor register maps, vendor-framed Modbus wrappers, and the Voltronic ASCII command protocol), and the register maps come overwhelmingly from community reverse engineering plus semi-public vendor PDFs.

One grounding paragraph on Modbus itself: Modbus is a 1979-era master/slave protocol where a client reads/writes numbered 16-bit registers using function codes (03 read holding registers, 04 read input registers, 06/16 write).

Modbus RTU runs over RS485 serial; Modbus TCP wraps the same PDU in TCP port 502. The protocol standardizes only the envelope — **which register means what is entirely vendor-defined**, which is exactly the "each company has its own pattern" problem from the notes. SunSpec (next section) is the attempt to standardize the register semantics.

Brand	Transport	Protocol	Register map source	Write/control documented?	Pakistan prevalence note
Sungrow	RS485 Modbus RTU; Modbus TCP via WiNet-S/LAN	Standard Modbus, Sungrow map	Official "Communication Protocol of Residential Hybrid Inverter" PDF circulates on forums, e.g. v1.1.11 PDF on the ioBroker forum and the openHAB thread sharing it	Yes — the mkaiser HA package controls "the EMS" incl. charge/discharge via documented holding registers	Strong in C&I/on-grid; Sungrow signed a 50 MW distribution deal at Solar Pakistan 2021
Growatt	RS485 Modbus RTU; ShineWiFi/Shine LAN dongle to Growatt cloud	Standard Modbus (RTU protocol doc versions v1.24/v1.39 referenced community-wide); dongle traffic is a proprietary TCP framing	Vendor PDF semi-public; register tables mirrored at e.g. AGG's Growatt Modbus map page; community repos like 0xAHA/Growatt_ModbusTCP (found via search, not opened). Dongle protocol reverse-engineered by grott	Partially — grott's "grottserver" can send read/write register commands (beta)	Very common in residential on-grid and hybrid in Pakistan (market claim **not verified** from a fetched page)
Deye (and Sunsynk/Sol-Ark rebadges)	RS485 Modbus RTU; Solarman logger on LAN	Standard Modbus, Deye map	Vendor "MODBUS RTU" PDF circulates, e.g. download.axilogi.com Deye MODBUS_RTU.pdf (found via search; not opened); community maps in kbialek/deye-inverter-mqtt and Solarman YAML profiles	Yes — deye-inverter-mqtt writes active power regulation (0–120%), solar sell, TOU schedules, work mode, battery charge params over MQTT, plus raw register write CLI (with damage warning)	Deye-based hybrids widely sold; Sunsynk is a Deye rebadge (README of deye-inverter-mqtt lists rebadged Bosswerk, Fuji Solar)
GoodWe	RS485/UDP; proprietary Aa55 + Modbus variants	Mixed proprietary/Modbus	Community: Home Assistant has a core GoodWe integration (local network protocol)	Core HA integration is read-oriented; write support **not verified**	Present via distributors (GoodWe 100 MW pact with Tradecorp — Indonesia-focused; PK share not verified)
Huawei (SUN2000)	RS485 Modbus RTU or Modbus TCP (port 502/6607) via SDongle	Standard Modbus, Huawei map; single-client limit	Community-documented via wlcrs/huawei_solar (AGPL-3.0, 939 stars on	Yes with "elevated permissions" (installer credentials): grid	Common in C&I and premium residential (prevalence **not verified**)

Brand	Transport	Protocol	Register map source	Write/control documented?	Pakistan prevalence note
			fetched page)	export control, forced battery charge/discharge	
Solis (Ginlong)	RS485 Modbus RTU; Solarman-type logger	Standard Modbus, Solis map	Community: ha-solarman profiles and wills106/homeassistant-solax-modbus covers Solis RHI models	Write via community integrations; official doc **not verified**	Solis signed a 300 MW distribution deal with AE Power for Pakistan, Mar 2024
Fox ESS	RS485 Modbus	Standard Modbus, Fox map	Community: nathanmarlor/foxess_modbus HA integration (unofficial, per its README banner)	Charge-period control exists in that integration (**not verified** in detail)	Growing presence; PK share not verified
Solax	RS485 Modbus / local API	Standard Modbus, Solax map	Community: wills106/homeassistant-solax-modbus (Apache-2.0, 528 stars on fetched page; Gen2–Gen5 hybrids) + HA core solax	Yes — wills106 README notes tested simultaneous read **and write** from two HA instances	Solax signed a 100 MW agreement with Fronus (Pakistani brand), Nov 2022 — suggests Fronus sells Solax-sourced units
Voltronic/Axpert family (MPP Solar, InfiniSolar…)	RS232/USB-HID, some RS485	**Proprietary ASCII command protocol (PI30/PI17/PI18)**: polled commands like `QPIGS` (status), `QPIRI` (ratings), CRC16-appended	Leaked/archived vendor docs, e.g. the Axpert RS232 protocol PDF on solarweb.net plus a decade of powerforum.co.za reverse engineering; implemented completely in jblance/mpp-solar	Yes — the protocol includes setting commands (output priority, charge current, voltages); community threads cover setting parameters on Axpert MKSII	**The most important family for Pakistan.** Many Pakistani-branded low-cost hybrids are widely held to be Voltronic rebadges (Inverex legacy lines, Knox, Crown, "Tesla" PK etc.) — we could not verify the specific rebadge list from a fetched page (w11stop blocked our fetch), so treat per-brand rebadge identity as **unverified**; SolarAssistant's own site says "many rebranded Voltronic or Deye inverters are also supported" (solar-assistant.io)
SRNE	RS485 Modbus RTU	Standard Modbus, SRNE map	Community: daniel-sullivan/srne-solar-	**Not verified**	SRNE-based units sold in PK low-cost

Brand	Transport	Protocol	Register map source	Write/control documented?	Pakistan prevalence note
			controller (Go library, found via search); SRNE listed WIP in wills106's integration; supported by SolarAssistant		segment (**not verified**)
LuxPower	RS485; proprietary LXP TCP packet via dongle	LXP packet protocol (framed Modbus-like)	Reverse engineered in celsworth/lxp-bridge (Rust, MIT, now unmaintained — points to jaredmauch/eg4-bridge)	lxp-bridge supports monitoring **and control** per its README	Luxpower listed as supported by SolarAssistant; PK share not verified

Two structural observations. First, the "pattern" differences are mostly **register-map differences over the same Modbus transport**, plus two genuinely different protocols (Voltronic ASCII; vendor dongle framings like Solarman V5, Growatt Shine, LXP). Second, the dongle problem from the notes is real and specific: the vendor Wi-Fi stick occupies the physical port. The escape hatch is that the dominant stick family — Solarman/IGEN loggers used by Deye, Solis, Sofar, and others — accepts Modbus RTU frames over its own LAN protocol: pysolarmanv5 documents the proprietary Solarman V5 framing on TCP port 8899 and sends arbitrary Modbus RTU requests through the stick, so the controller can piggyback on the dongle instead of displacing it. Similarly, grott sits as a man-in-the-middle between a Growatt ShineWiFi stick and Growatt's cloud, parsing (and in beta, writing) without touching the serial port.

Standards

The point: SunSpec Modbus is the only serious cross-vendor register standard, but the Asian residential/hybrid brands that dominate Pakistan largely ignore it, so a product cannot rely on it — it can only borrow its data model.

SunSpec Modbus standardizes "information models" — fixed blocks of registers with defined meanings — discovered by scanning for the "SunS" marker at base addresses 0, 40000 or 50000, as implemented in the Alliance's own pysunspec2 Python library (Apache-2.0, 74 stars on the fetched page; models in SMDX/JSON; `scan()` then read/write Points). The newer 700-series DER models were approved in 2021 to carry IEEE 1547-2018 functions (volt-var, freq-watt, limit-watt). SunSpec matters in North America because IEEE 1547-2018 requires inverters to support one of SunSpec Modbus, IEEE 2030.5, or IEEE 1815, as explained by Solar Builder; California's Rule 21/CSIP stack uses IEEE 2030.5. SunSpec maintains a product certification registry (per-vendor pages; Alpha ESS shown; a full machine-readable list was not fetchable in this pass), and vendors like SMA publish SunSpec Modbus profiles. Some Chinese vendors are members — Sofar is a SunSpec contributing member — but membership is not implementation.

Why the Pakistani mix doesn't comply: the US/EU interconnection rules that force compliance don't apply in Pakistan; the hybrid/off-grid segment that dominates there evolved from UPS lineage (Voltronic's ASCII protocol predates SunSpec adoption in that class); and vendors have a commercial incentive to keep customers inside their dongle+cloud ecosystem. Practical consequence for OnyxLeaf: **treat SunSpec's data model (models 103/802/7xx etc.) as the internal normalized schema**, map every proprietary device onto it, and get SunSpec-compliant devices nearly free — but never assume a field device speaks it.

Open-source codebases

The point: the "driver library" the meeting notes imagine already exists in open source to a remarkable degree — read-side coverage of essentially every brand above, and write-side coverage for Deye, Sungrow, Huawei, Solax,

Voltronic and LuxPower — but it is fragmented across ~10 projects, three languages, and licenses ranging from MIT to AGPL.

Repo	Language	License	Scale/activity (from fetched page)	Covers	Read/Write	Maps defined as
jblance/mpp-solar	Python	MIT	438★, 1,848 commits	Voltronic/ MPP family (PIP, Axpert, LV5048), also JK/Daly BMS, Victron VE.Direct	Read + settings commands (protocol includes set commands; page emphasizes read)	Command/ response definitions in Python package
johanmeijer/ grott	Python	not shown on fetched page	534★, 351 commits, v2.8.3 (2024-05-17)	Growatt via Shine dongle MITM (proxy or sniffer)	Read; write via grottserver (beta); `blockcmd` safety filter	Python record-layout parsing
bohdan-s/ SunGather	Python 3	GPL-3.0	214★, 257 commits	Sungrow SG/SH ranges via Modbus TCP (WiNet-S)	Read only	YAML register files from official Sungrow docs; auto-detects model
mkaiser/ Sungrow-SHx-...-Home-Assistant	HA YAML	MIT	708★, 561 commits, active Jan 2026	Sungrow SH hybrids	Read + **EMS control** (charge/discharge tab)	Pure YAML Modbus config — a worked example of maps-as-data
StephanJoubert/ home_assistant_solarman	Python	Apache-2.0	741★, archived — points to davidrapan	Deye/ Sunsynk/Sol-Ark, Sofar, Solis, ZCS via Solarman V5 logger	Read (service hook for advanced use)	**YAML `inverter_definitions` per device — the closest existing thing to a community register-map database**
davidrapan/ ha-solarman	Python	MIT	516★, 1,191 commits	Successor; Deye, Solis, Sofar, Solax, Sunsynk (topics)	Read; write capability not stated on fetched page	YAML profiles (wiki-documented)
jmccrohan/ pysolarmanv5	Python	MIT	180★, 268 commits	Solarman V5 logger framing (TCP 8899) for any Modbus RTU payload	Arbitrary Modbus requests incl. writes (write not explicitly stated on fetched page)	Library, not maps
kbialek/deye-inverter-mqtt	Python	Apache-2.0	364★, 825 commits	Deye string/micro/ hybrid + rebadges (Bosswerk, Fuji)	Read + **write**: active power regulation, TOU schedules, work mode, raw register CLI	Metric groups with per-group register docs

Repo	Language	License	Scale/activity (from fetched page)	Covers	Read/Write	Maps defined as
OpenEMS/openems	Java	**EPL-2.0 (Edge/Backend), AGPL-3.0 (UI)**	~1.6k★, 6,722 commits; founded by FENECON, OpenEMS Association, EU/German funding	Batteries (BYD, Pylontech, Soltaro…), battery inverters (Kaco, REFU, Sinexcel), bridges for Modbus/MBus/MQTT/OneWire	Read + control (it is an EMS)	Java "Nature"/Channel abstraction: devices implement Natures (e.g. SymmetricEss) exposing typed Channels; `io.openems.edge.bridge.modbus` maps registers→channels. The architecture to copy; details on fetched page were thin — deep-dive its docs next
evcc-io/evcc	Go (+Vue UI)	MIT	~7.5k★, 10,236 commits	Dozens of vendors incl. Sungrow, Huawei, GoodWe, Victron, generic SunSpec	Read + battery control (feature set on docs.evcc.io; not detailed on fetched page)	Top-level `templates/` folder of device definitions (YAML; format not shown on fetched page)
Victron Venus OS + mr-manuel/venus-os_dbus-serialbattery	Python	AGPL-3.0	276★, 1,045 commits	Many BMS over RS232/RS485/TTL/BLE → Victron dbus	Read BMS; **steers charging** by publishing `/Info/MaxChargeCurrent`, `MaxDischargeCurrent`, `MaxChargeVoltage` to dbus	Per-BMS Python `Battery` classes; auto-detects by trying each class's connection test — a working example of probe-and-identify
meltaxa/solariot	Python	(not fetched in detail)	found via search	Sungrow/SMA Modbus→MQTT	Read	Python register maps; relevant as prior art only
modbus4mqtt	Python	(PyPI page)	found via search	Generic	"YAML-defined **bidirectional** Modbus to MQTT" per its PyPI description	YAML
wlcrs/huawei_solar	Python	AGPL-3.0	939★, 497 commits	Huawei SUN2000 + batteries + meters	Read + write with "elevate permissions" (installer creds): export control,	Python register definitions (huawei-solar lib)

Repo	Language	License	Scale/activity (from fetched page)	Covers	Read/Write	Maps defined as
					forced charge/discharge	
wills106/homeassistant-solax-modbus	Python	Apache-2.0	528★, 3,848 commits	Solax Gen2–5, Sofar, Solis, Growatt, + WIP SRNE/Solinteg/Swatten	Read + write (multi-instance read/write noted)	Python per-brand plugin files
nathanmarlor/foxess_modbus	Python	(page found via search)	—	Fox ESS	Read + charge windows (**not verified** in detail)	Python
celsworth/lxp-bridge	Rust	MIT	40★, 349 commits, **unmaintained** → eg4-bridge	LuxPower LXP dongle protocol	Read + control	Rust
syssi/esphome-jk-bms	ESPHome/C++	Apache-2.0	~1.0k★, 522 commits, requires ESPHome ≥2026.1	JK BMS via UART-TTL/BLE	Read + some control (balancer/charge/discharge switches)	ESPHome YAML + C++ component
ArminJo/JK-BMSToPylontechCAN	Arduino C++	(found via search)	—	Translates JK-BMS data to **Pylontech CAN** so any inverter that "speaks Pylontech" accepts it	Emulation/write	Code

Per-repo notes beyond the table. **mpp-solar** is the de-facto reference implementation of the Voltronic family and therefore the single most Pakistan-relevant repo; it is MIT, pure Python, and structured as a library of commands/responses — OnyxLeaf can embed it or port its protocol tables outright. **grott** demonstrates the dongle-MITM pattern: don't fight the vendor stick, impersonate its server; its `blockcmd` option (blocking reboot/configure commands from outside) is a fail-safe idea worth copying. **SunGather vs mkaiser**: SunGather (GPL-3.0 — copyleft, caution) proves Sungrow model auto-detection from Modbus; mkaiser proves full hybrid control (EMS mode, forced charge/discharge) with nothing but YAML over the official Sungrow register PDF. **The Solarman pair** (StephanJoubert archived → davidrapan active, on jmccrohan's pysolarmanv5) is the architectural highlight: a generic transport library + a folder of YAML `inverter_definitions`, one per device family, selected by a `lookup_file` — this is the community register-map database the meeting notes wished existed, and it is Apache/MIT licensed. **kbialek/deye-inverter-mqtt** is the best evidence that cross-brand *control* is real: documented MQTT topics write Deye active-power regulation (0–120%), TOU schedules and work modes, with an explicit warning that invalid raw writes "may damage the inverter". **OpenEMS** is the most mature architecture to study (not necessarily adopt — Java/OSGi is heavy for a cheap edge box): devices implement abstract "Natures" with typed Channels, control algorithms are written against Natures and thus device-independent, and a Modbus bridge maps registers to channels; it also proves the business model (FENECON built its commercial FEMS on it). **evcc** proves the same maps-as-data idea at scale in Go with a `templates/` folder and MIT licensing. **dbus-serialbattery** is the de-facto open battery-integration library and a working auto-detect example (it tries each BMS class's connection test until one succeeds); it is AGPL-3.0, so reuse means ideas, not code, unless OnyxLeaf open-sources that component.

License/reuse analysis. For a commercial embedded controller + proprietary cloud: (a) **MIT/Apache code is freely reusable** with attribution — that covers mpp-solar, pysolarmanv5, davidrapan/ha-solarman, mkaiser's YAML, kbialek (Apache-2.0), wills106 (Apache-2.0), evcc (MIT), modbus4mqtt. (b) **GPL-3.0 (SunGather)**: shipping it on the box

obliges releasing the combined work's source to customers; avoid linking, reimplement from the underlying Sungrow PDF instead. (c) **AGPL-3.0 (wlcrs/huawei_solar, dbus-serialbattery, OpenEMS UI)**: AGPL's network clause makes it toxic for a closed cloud; use as documentation of register semantics, write clean implementations. (d) **OpenEMS Edge/Backend is EPL-2.0**, which is file-level copyleft and generally compatible with commercial use if modifications to EPL files are published — viable if OnyxLeaf wanted to build *on* OpenEMS rather than beside it. (e) Critically, **register maps as facts are not copyrightable in most jurisdictions, but the YAML files expressing them are**; Apache/MIT-licensed YAML profiles can be imported wholesale with attribution, which bootstraps OnyxLeaf's driver library on day one. Roughly: read-drivers for every target brand and write-drivers for 5–6 brands already exist under permissive licenses; OnyxLeaf's net-new work is the unified schema, the identification layer, the fail-safe control layer, the Voltronic-rebadge QA matrix for Pakistan, and productization.

Pattern detection feasibility

Verdict first: **"detect all patterns and adapt" is feasible — but as curated-library fingerprinting, not as ML or automatic protocol inference.** No shipping multi-brand product learns protocols; all of them select from a human-built driver library.

(a) How existing tools actually do it. SolarAssistant supports ~16 brand families and "many rebranded Voltronic or Deye inverters" via a compatibility guide — i.e., a curated driver set with user/auto selection. SunGather "auto-detects the inverter model on first connection and selects the matching registers" — it reads Sungrow's device-type register and picks a known map. dbus-serialbattery "tries each known Battery implementation's connection test and keeps the one that succeeds" — brute-force probing over a driver library. ha-solarman autodiscovers the logger on the LAN and selects a YAML profile. evcc and OpenEMS make the installer pick a template/component, then validate. The pattern is universal: **transport probe → identity read → map selection → validation.**

(b) Automatic protocol reverse engineering is a research field, not a product technique. Netzob ("Protocol Reverse Engineering, Modeling and Fuzzing", per its PyPI description) infers message formats from captured traffic by sequence alignment and clustering — useful offline, in the lab, to accelerate a human reverse-engineering a new dongle protocol. It cannot tell you that register 40087 is "battery SoC in 0.1%" vs "grid frequency in 0.01 Hz"; semantics require a human with the device and a ground truth. For *control*, automatic inference is disqualifying: a mis-inferred write register on a 10 kW hybrid is a fire/warranty/legal event. The honest line for investors: ML cannot conjure register semantics; humans curate maps, software selects them.

(c) The realistic method, concretely:

1. **Transport scan**: enumerate serial ports and LAN (Solarman port 8899 scan — pysolarmanv5 ships a network scanner; Modbus TCP 502; Growatt Shine). Try baud rates 9600/115200, Modbus slave IDs 1–5, and Voltronic `QPI`/`QPIGS` probes.
2. **Identity read**: each family has cheap identity probes — Sungrow device-type code register, Deye/Solis serial-number registers via Solarman, Voltronic `QPI` returns the protocol ID (`PI30` etc.), SunSpec "SunS" scan at 0/40000/50000 (pysunspec2 behavior).
3. **Map scoring**: load candidate YAML maps for that family; read 15–30 registers; score each map by plausibility constraints (voltage 150–280 V AC, frequency 49.5–50.5 Hz, SoC 0–100%, power ≤ rated). Accept the best map above a threshold; otherwise flag "unknown device" and upload a raw register dump for human curation. This scoring step is the only legitimate "pattern detection" automation, and it is a weekend of code, not a research program.
4. **Declarative register-map DSL**: every device is a YAML file (registers, scaling, enums, write points with min/max/step and safety class), versioned in a repo, hot-loadable — exactly the Solarman `inverter_definitions` / evcc templates model. New devices become data, shippable OTA without firmware releases.
5. **Conformance tests**: every map ships with recorded request/response fixtures and a bench checklist; write points are gated until bench-verified on real hardware.

What ML can honestly add: anomaly detection on normalized telemetry (drifting SoC, clipping, failing fans), fleet-level fault classification, and perhaps assisted map curation (clustering raw dumps from unknown devices against known families). That plays to OnyxLeaf's Boxlytics "predict and optimise" DNA — but it sits on top of the data layer, not inside protocol handling.

Battery data

The wrong-SoC problem is real and structural: low-cost rebadged LFP packs ship BMS firmware whose SoC is naive coulomb counting that drifts (LFP's flat open-circuit-voltage curve between ~20–80% SoC makes voltage-based correction hard, so cheap BMSs rarely recalibrate except at full charge), and the inverter blindly displays whatever the BMS reports over the Pylontech-compatible protocol. Community tooling confirms both halves: the Pylontech RS485 console protocol has been publicly dissected (e.g. the EEVblog "Pylontech sc0500 protocol hacking" thread), "Pylontech CAN" has become the lingua franca batteries use to talk to hybrid inverters — to the point that ArminJo/JK-BMSToPylontechCAN exists purely to make a JK BMS *impersonate* a Pylontech on CAN so any inverter accepts it — and dbus-serialbattery exists largely because stock BMS data is not good enough for Victron systems, publishing its own charge/discharge limits to steer the inverter.

Estimator approach for OnyxLeaf: the controller already logs pack voltage, current and temperature at 1–10 s resolution from both the inverter and (where reachable) the BMS directly (RS485/CAN; JK via syssi/esphome-jk-bms-style components). On that stream, run a software SoC estimator: coulomb counting as the fast integrator, OCV-based correction applied only in the usable OCV regions (rest periods, and the curve's steep ends), fused by an extended Kalman filter; the method triad (OCV vs coulomb counting vs Kalman) is standard BMS practice as summarized in e.g. Sunlith's SOC estimation overview, with the EKF literature represented by open-access treatments such as this Energies 2022 EKF SoC paper. PyBaMM — correction: PyBaMM was **not fetched** in this pass; cite it as the standard open-source battery modeling framework (**not verified** here) — can generate OCV/parameter priors per cell chemistry offline. Realistic accuracy with V/I/T only and periodic full-charge resets is ±3–5% SoC mid-range for LFP (**engineering estimate, not a verified benchmark**) — which is a dramatic improvement over packs that display 60% and then cut off, and also yields SoH trending (capacity fade from charge-throughput between full-charge events) as a differentiating feature. The product displays *both* numbers ("BMS says 72%, estimated 58%") to build installer trust rather than silently overriding.

Hardware

The point: a Raspberry-Pi-class Linux box with 2× isolated RS485 and 1× CAN covers every target device; this is a $60–120 BOM item, and SolarAssistant proves customers accept Pi hardware at a $149 retail price point.

Option	What	Approx. cost	Notes
Raspberry Pi Zero 2 W kit	What SolarAssistant ships: Pi Zero 2 W + Ethernet/USB board + 32 GB Samsung EVO card + enclosure + 9–60 V DC supply option, $149 retail incl. software	$149 retail (bundle)	Proves the form factor and price anchor; DC supply off the battery bus is a good idea to copy
Raspberry Pi CM4/CM5 on custom carrier	SoM + carrier with 2× isolated RS485, 1× CAN, Ethernet, 4G module option	CM4 from ~$30–45 (SoM, list; street prices vary — **not verified** this pass) + carrier	The scalable path: industrial temp range option, eMMC (SD cards die in Pakistani heat), custom carrier is OnyxLeaf IP
Pi 4/5 + Waveshare RS485 CAN HAT	Off-the-shelf HAT gives RS485 + CAN	HAT ~$15–20 at resellers (e.g. ThePiHut listing); exact price **not verified**	Fastest bench/pilot rig; not ruggedized
Industrial RS485-to-Ethernet/Wi-Fi bridges	Waveshare/EW11-class converters put the inverter's RS485 on the LAN; a how-to for Growatt	~$20–40 each (**not verified**)	Lets one central controller serve multiple inverters; good for C&I retrofits

Option	What	Approx. cost	Notes
	via EW11 exists; Wattuneed sells a Deye+Waveshare gateway bundle		
ESP32-class	ESPHome-style firmware per device (syssi/esphome-jk-bms pattern)	$5–10	Too small for the multi-protocol controller itself, but ideal as cheap per-device probes feeding the main box
Vendor dongle piggyback (no new port hardware)	Use the existing Solarman stick via pysolarmanv5 over TCP 8899, or MITM Growatt Shine via grott	$0 marginal	Solves the "dongle occupies the port" objection; LAN only, vendor can break it with firmware updates

On Solarman/IGEN: the sticks are made by IGEN Tech, whose SOLARMAN platform is a white-label logger+cloud business sold to inverter OEMs and distributors (SOLARMAN Business) — i.e., many "vendor clouds" in this market are actually one Chinese platform company, which is both a competitive precedent (vendor-agnostic data businesses work) and a vulnerability of the incumbents (their "lock-in" is outsourced).

Commercial landscape

The point: a direct price anchor exists at $149 one-time (SolarAssistant), the C&I end is served by quote-priced plant controllers (Elum), and the named Pakistani channel partners could not be verified online — that must be resolved by asking the engineering manager for exact names/contacts.

Player	What they sell	Pricing	Brands/coverage	Relevance
SolarAssistant	Pi-based local monitoring + remote settings adjustment	One-time, per-device, perpetual license; free updates; $149 hardware+software bundle; software-only license price not captured (**not verified**)	EG4, Sol-Ark, Growatt, MPP Solar, SunGoldPower, SRNE, Deye, Voltronic, Luxpower, Victron, PowMr, Vevor + "many rebranded Voltronic or Deye"	The closest competitor; strong in South Africa/US off-grid; explicitly markets "remotely adjust inverter settings" to installers. No Pakistan presence mentioned
Elum Energy	ePowerControl plant controller, ePowerMonitor, eConf	Quote only (RFI form)	"Interface with any power asset"; compatibility checker; 90+ countries claimed; Paris/Casablanca offices	Proves vendor-agnostic C&I controllers are a real paid category; Elum also publishes "top hybrid inverter" rankings. Pakistan not mentioned on fetched page
IGEN Tech / SOLARMAN	OEM dataloggers + white-label monitoring cloud for brands/distributors	B2B/OEM	Hundreds of OEMs (claimed)	The incumbent "vendor cloud" is often actually Solarman; their V5 protocol is open-ish via pysolarmanv5
Victron GX ecosystem	Cerbo GX etc. running open Venus OS (dbus architecture) with huge third-party driver community (dbus-serialbattery)	Hardware margin	Victron-centric but famously open	The model for "open edge OS + paid hardware"; premium segment in PK
OpenEMS / FENECON	Open EMS platform; FENECON's commercial FEMS	Open core	German/EU C&I	Proof of open-core commercial model

Player	What they sell	Pricing	Brands/coverage	Relevance
	built on it			
ComAp, Studer	Genset/hybrid controllers (ComAp) and off-grid inverter control (Studer)	—	—	**Not fetched this pass**; known category players, include in later deep dive (not verified)
Pakistani channel: EnerCon, Greenfuels, NeoSat Renewable	Per the meeting notes: sell controllers/hardware in Pakistan	—	—	**Could not be verified online.** `enercon.com.pk` and `greenfuels.pk` both refused fetching (robots disallowed), so their businesses are unconfirmed; searches for "NeoSat Renewable Pakistan" returned no matching company (only unrelated Sungrow/Solis distribution news, e.g. Solis–AE Power 300 MW). Action: get exact spellings/contacts from the engineering manager

Who buys vendor-agnostic controllers today: off-grid/hybrid *owners and installers* (SolarAssistant's market — tens of dollars to low hundreds, one-time), *C&I EPCs and operators* (Elum/ComAp class — quote-priced, thousands per site), and *OEM/distributors* who white-label monitoring (Solarman's market). Pakistan's load-shedding-driven hybrid boom maps onto the first segment at scale and the second in textile/industrial C&I.

Control-write risk

The point: write access is the product's differentiator and its biggest liability; evidence says control is possible on the major brands but must be engineered as a safety system, not a feature flag.

Findings. Vendors differ sharply in how sanctioned third-party control is: **Huawei** gates write access behind "elevate permissions" with installer-account credentials, after which forced battery charge/discharge and export control work (wlcrs/huawei_solar) — i.e., officially contemplated but credentialed. **Sungrow** publishes its hybrid register map including EMS-mode and charge/discharge registers, which the mkaiser YAML package drives in production in thousands of homes (708★), with the maintainer still flagging untested functions. **Deye** writes are community-routine but explicitly at-your-own-risk: kbialek's README warns invalid values "may damage the inverter", and forum threads like powerforum's "how to safely change settings on a Deye inverter" and German threads on Deye 12K/20K active-power setpoints via Modbus show both demand and nervousness. **Voltronic** setting commands are long-established (parameter-setting threads such as Axpert MKSII settings via protocol), and SolarAssistant commercially ships remote settings adjustment on these families (solar-assistant.io). We did not find (in this pass) a documented case of a warranty formally voided for Modbus writes — absence of evidence, **not verified** either way — but vendor terms generally reserve that right, and grott shipping a `blockcmd` feature specifically to block remote configure/reboot commands shows the community treats stray writes as a real hazard. A further practical risk from the Huawei page: Modbus often allows **one client connection at a time**, so the controller must own the port and multiplex all consumers.

Fail-safe design (to put in the spec): (1) **write allow-list per device map** — only bench-verified registers, with min/max/step/rate limits encoded in the YAML, everything else read-only; (2) **watchdog/dead-man pattern** — every

external setpoint has a TTL; on controller crash, comms loss, or cloud loss, the inverter is reverted to its own standalone mode (self-consumption/TOU as configured locally), mirroring how dbus-serialbattery steers via published limits the inverter can ignore rather than seizing control; (3) **read-back verification** after every write, alarm on mismatch; (4) **two-tier product**: monitoring-only mode by default, control unlocked per-site after a conformance check, with per-brand legal language; (5) **command audit log** shipped to cloud for dispute defense.

Product scope and methodology

The point: build a thin, boring, data-driven edge stack whose entire brand knowledge lives in versioned YAML maps, and spend the engineering budget on identification, fail-safety, and the bench.

Driver abstraction. Internal normalized model = a SunSpec-shaped subset (inverter AC/DC telemetry ≈ models 101–103, storage ≈ 802, controls ≈ 7xx semantics) implemented as typed channels, borrowing OpenEMS's Nature/Channel idea (OpenEMS) without the Java stack — Go or Python on Linux. Three transport plugins (Modbus RTU/TCP incl. Solarman V5 framing; Voltronic ASCII per mpp-solar; vendor-cloud MITM per grott) + CAN for batteries. On top, a **register-map DSL in YAML**: device identity probes, read blocks with scaling/enums, write points with safety metadata — seeded by converting the Apache/MIT-licensed community maps (ha-solarman definitions, mkaiser Sungrow YAML, kbialek Deye docs, wills106 plugins) with attribution. **Probe-and-identify flow** as specified in §Pattern detection: transport scan → identity read → candidate-map scoring with plausibility constraints → accept/flag. **Conformance rig**: a bench with one real unit of each target brand; every map version must pass recorded-fixture replay + live read sanity + (for write points) a scripted write/read-back/revert cycle on the bench before it can be enabled in the field; unknown field devices upload raw dumps that become new maps centrally.

Build order for Pakistan (first 5 driver families): 1) **Voltronic/Axpert family** (covers Inverex-class local rebadges — the volume segment; MIT-licensed reference in mpp-solar); 2) **Deye family** (incl. Sunsynk rebadges; strong write support precedent); 3) **Growatt** (serial Modbus + Shine MITM); 4) **Sungrow** (C&I credibility; official PDF + proven control YAML); 5) **Solis via Solarman** (fresh 300 MW PK distribution), with the Solarman V5 transport also unlocking Sofar/others nearly free. Battery side in parallel: Pylontech-CAN-compatible reading (the de-facto standard) + JK BMS, then the SoC estimator.

Edge/cloud split. Edge box (CM4-class, 2× isolated RS485 + CAN, eMMC, DC-input PSU off the battery bus like SolarAssistant's option A): all protocol work, local buffering for Pakistan's connectivity gaps, local web UI, watchdog-governed control. Cloud: fleet dashboard, map distribution/OTA, alerting, the normalized API that Product 1 (EMS/optimizer) consumes, audit logs. The controller must work indefinitely offline — local-first is both a reliability requirement and the anti-lock-in marketing story.

Validation and pilot plan

The point: validate maps on a bench you own before any field write, then let 5 real mixed-brand sites falsify the coverage claim.

Phase 1 — bench rig (weeks 1–8): acquire 1 unit each of the first 5 families (used/local market; include at least one Pakistani-brand Voltronic rebadge and one Deye rebadge) + 2 battery types (Pylontech-compatible rack LFP, JK-BMS DIY pack). Deliverables: identification success on all units from cold start; read conformance (every normalized channel within plausibility bounds vs front-panel values); write conformance on the allow-listed setpoints with revert; 72-hour soak without comms lockups (watch the single-Modbus-client constraint noted on wlcrs/huawei_solar).

Phase 2 — site surveys (weeks 6–12, overlapping): 5 sites sourced through the channel (once EnerCon/Greenfuels/NeoSat contacts are confirmed — currently unverified) spanning residential hybrid, small C&I, and one mixed-brand site. At each: inventory exact models, dongles present, ports free/occupied, battery brand and whether displayed SoC is believed; install in monitoring-only mode; log for 4 weeks.

Success metrics: ≥90% of encountered devices auto-identified without manual map work; ≥95% telemetry completeness over 4 weeks; SoC estimator vs full-charge-reset ground truth within ±5%; at least 3 of 5 brands accept

≥1 safe write (charge-current limit or TOU window) with clean read-back; one installer willing to pay a stated price for 10 more units (the real metric).

Pricing, risks and kill criteria

Pricing anchors. SolarAssistant: one-time perpetual per-device license, [90–160) for residential/installer, plus an optional fleet-cloud subscription for installers (per-site/month) — subscription on the cloud, never on the box, to keep the anti-lock-in story coherent. C&I controller-with-control is a different SKU at 5–10× via the Elum-style quote model (elum-energy.com).

Risks: (1) vendor dongle/firmware updates breaking MITM and logger piggybacks (mitigate: direct-RS485 install option always available); (2) GPL/AGPL contamination if community code is vendored carelessly (mitigate: license audit gate in CI; reimplement from PDFs where needed); (3) SolarAssistant entering Pakistan or local clones undercutting (mitigate: control + battery-truth + Urdu/installer channel as moats); (4) damage/warranty liability from writes (mitigate: §Control-write fail-safes, insurance, per-brand disclaimers); (5) the channel partners being installers who want margin on hardware, not software partners (unknown until verified); (6) grey-market model churn: Pakistani rebadges change OEM internals without changing the shell, so identification must trust protocol probes, never the logo.

Kill criteria: kill or descope to monitoring-only if, on the bench, **fewer than 3 of the first 5 brand families expose any safely writable control register/command** (the notes' core promise of "control for any brand" would then be false in this market); kill the auto-detect pitch if >30% of pilot-site devices need manual map work after the first 5 families are built; kill the battery-estimator feature if V/I/T-only accuracy cannot beat the OEM BMS error on the bench; reconsider the whole product if pilot installers won't pay more than commodity-dongle prices (≤$30) for it.

Saudi Arabia and GCC

The point: the GCC is a different product on the same platform — tier-1 brands (Huawei/Sungrow/SMA class) with real Modbus/SunSpec support replace the Voltronic rebadge problem, but regulated equipment lists, SASO/DEWA certification, and accredited-installer regimes make it a year-2 market best entered through ESCO/O&M and hybrid-plant partners rather than direct retail.

Brand mix. KSA's solar inverter market is led by global tier-1 vendors: a ResearchAndMarkets outlook names Fronius, Sungrow, SMA, Schneider Electric, ABB and Huawei Solar Arabia as key players in Saudi Arabia (2022–2028 report; exact shares behind paywall, **not verified**), consistent with Huawei and Sungrow leading Wood Mackenzie's global inverter ranking for H1 2025. Dubai's eligible-equipment regime has historically listed European/tier-1 units — Ingeteam, REFUsol, Schneider Electric and more recently Solplanet all announced DEWA listing. Consequence for the roadmap: the Pakistan first-5 (Voltronic, Deye, Growatt, Sungrow, Solis) does **not** transfer — a GCC driver set is Huawei, Sungrow, SMA, GoodWe/Growatt (C&I/resi), plus SunSpec-generic, and three of those (Sungrow, Huawei, SunSpec) are already in the plan or nearly free. Voltronic-family work is irrelevant in GCC grid-tied segments.

Regulation and certification. UAE/Dubai: DEWA's Shams Dubai programme requires both panels and inverters to appear on DEWA's approved equipment list (150+ components, via the Hab Reeh platform), with inverters certified to IEC 62109-1/-2 and IEC 62116 anti-islanding, and every application submitted by a DEWA-accredited consultant/contractor (111 active in 2025); DEWA's own eligible-equipment page returned an error when fetched (list contents **not verified** directly), and an older 2017 DRRG equipment list PDF exists on dewa.gov.ae (not opened). The fetched guide states **no monitoring/communication interface mandate** — no SunSpec/IEC 61850 requirement appears for small-scale DRRG systems — and DEWA simply installs a bidirectional smart meter. KSA: the ECRA Regulatory Framework for Small-Scale Solar PV (ERD-TA-012) covers 1 kW–2 MW systems, requires "material compliance with Saudi standards or equivalent international standards" with the DSP and SASO maintaining an approved-components list, mandates certified installers, requires a separate generation meter only above 100 kW, and **contains no provision for remote/third-party control and no communication-interface specification** (ECRA has since been renamed; current SERA-era rules **not verified** this pass). For the controller as an imported electronic product, KSA requires a SASO Certificate of Conformity, and specified electronic devices additionally need a SASO

IECEE recognition certificate issued after CB-scheme lab testing, valid 1 or 3 years, with Arabic manuals and a CITC certificate if the device has SIM/data connectivity — a 4G-equipped controller would trigger the CITC requirement; costs/timelines were not stated (**not verified**; budget assumption: low thousands of USD and 2–4 months per the certification industry's norms, unverified). Net regulatory read: nothing we fetched *mandates* SunSpec or forbids owner-side control of inverters behind the meter, but the controller must not interfere with mandated anti-islanding/protection behavior, and the accredited-installer regime means the channel must be licensed firms, not end-users.

Demand side. The vendor-agnostic control use case is strongest in GCC solar-diesel hybrids and mixed-brand O&M portfolios. Enerwhere (Dubai) runs rental solar-diesel microgrids for construction and oil-and-gas camps, e.g. a 2 MVA PPA-based hybrid at the Buhasa workforce camp in Abu Dhabi, and deployed the Middle East's then-largest Li-ion battery on a DIFC construction site — exactly the customer profile that needs one controller across whatever inverters and gensets each rental site has. Oman's Tanweer (Rural Areas Electricity Co.) has planned ~200 MW of solar-diesel hybrids and tendered 11 hybrid PV-diesel sites totaling ~160 MW — utility-grade, controller-mandatory projects (won by large IPPs; OnyxLeaf relevance is as a subcomponent vendor at best, **speculative**). On the closest commercial analogue: Elum's ePowerControl compatibility tool exists but rendered empty on fetch (brand list **not verified**), its references page claims deployments in "more than 50 countries" without naming them, and we found an Arabic-language report of Marizad taking an exclusive Elum agency (region/date **not verified**) — so direct evidence of GCC ePowerControl installs was not obtained, though its solar-diesel positioning targets exactly this market.

Conclusion for the roadmap. GCC entry is a year-2 move: (1) swap the driver priority to Huawei/Sungrow/SMA/GoodWe + SunSpec-generic (two of five shared with Pakistan; SMA is largely SunSpec, cheap to add); (2) budget SASO CoC + IECEE (and CITC if cellular) for KSA and plan DEWA/accredited-installer channel partnerships for Dubai rather than any consumer channel; (3) lead with the solar-diesel-hybrid and multi-brand O&M segments (Enerwhere-class ESCOs, camp operators, O&M contractors), where vendor-agnostic control is already a purchased category and where Pakistan-honed genset-plus-PV economics transfer; (4) do not carry the Voltronic/battery-SoC story to GCC marketing — the wedge there is fleet normalization and fuel-save control, not wrong battery data.

Open questions

1. Exact identities/contacts of "EnerCon", "Greenfuels", "NeoSat Renewable" — neither of the obvious domains was fetchable (robots disallowed) and "NeoSat Renewable" returned no match; are the names spelled differently, and are they distributors, installers, or importers?
2. Which Pakistani brands are currently Voltronic rebadges vs Deye/SRNE/Megarevo rebadges, model-by-model (Inverex Nitrox lines appear to include non-Voltronic OEM units; needs teardown-level confirmation).
3. Real brand mix by installed base in Punjab/Islamabad residential vs C&I — no credible public market-share data found; the 5-site survey plus installer interviews must substitute.
4. Growatt serial write registers (not via Shine MITM): which protocol doc version (v1.24 vs v1.39 vs newer) matches the SPF/hybrid units sold in PK?
5. NEPRA/net-metering regulatory posture on third-party export-limit control, and whether any PK DISCO requires vendor-certified controllers.
6. SolarAssistant software-only license price and whether it is sold/used in Pakistan today.
7. Whether Solarman stick firmware in PK-market units still accepts local port-8899 V5 connections (newer sticks reportedly restrict this — **unverified**).
8. Legal review: enforceability of warranty-void claims for register writes under Pakistani consumer law.

Sources

Open-source repositories (fetched):

- jblance/mpp-solar · johanmeijer/grott · bohdan-s/SunGather · mkaiser/Sungrow-SHx-Inverter-Modbus-Home-Assistant

- StephanJoubert/home_assistant_solarman · davidrapan/ha-solarman · jmccrohan/pysolarmanv5 · kbialek/deye-inverter-mqtt
- OpenEMS/openems · evcc-io/evcc · mr-manuel/venus-os_dbus-serialbattery · syssi/esphome-jk-bms
- sunspec/pysunspec2 · wlcrs/huawei_solar · wills106/homeassistant-solax-modbus · celsworth/lxp-bridge

Repos/libraries found via search (pages not fully opened): meltaxa/solariot · modbus4mqtt (PyPI) · nathanmarlor/foxess_modbus · 0xAHA/Growatt_ModbusTCP · daniel-sullivan/srne-solar-controller · ArminJo/JK-BMSToPylontechCAN · AMOSSYS/Netzob / Netzob on PyPI

Protocol documents and threads: Sungrow residential hybrid protocol v1.1.11 PDF (ioBroker forum mirror) · openHAB Sungrow protocol thread · AGG Growatt Modbus map · Deye MODBUS_RTU.pdf (axilogi mirror) · Voltronic Axpert RS232 protocol PDF · powerforum Voltronic USB reading · powerforum Axpert settings · powerforum Deye safe settings · akkudoktor Deye Modbus setpoints · EEVblog Pylontech protocol hacking

Standards: SunSpec specifications · 700-series approval · SunSpec certification registry (example page) · Sofar membership · SMA SunSpec profile PDF · Solar Builder on IEEE 1547/Modbus

Commercial: SolarAssistant home · SolarAssistant licensing update · SolarAssistant Pi Zero device, $149 · Elum Energy · Elum hybrid-inverter ranking · SOLARMAN Business · SOLARMAN Smart brochure PDF · Wattuneed Deye RS485 gateway · EW11 Growatt bridge how-to

Pakistan market: Sungrow 50 MW PK distribution · Solis–AE Power 300 MW PK · Solax–Fronus 100 MW · GoodWe–Tradecorp 100 MW · enercon.com.pk and greenfuels.pk — fetch refused (robots disallowed), contents unverified

Battery/estimation: Sunlith SOC methods overview · Energies 2022 EKF SoC paper (DOAJ) · HA GoodWe integration · HA SolaX integration · HA growatt_server

Hardware: Waveshare RS485 CAN HAT · ThePiHut RS485 CAN HAT listing

Saudi Arabia and GCC: Saudi solar inverter market key players (Businesswire/ResearchAndMarkets) · Huawei/Sungrow lead H1 2025 inverter rankings (SolarQuarter/Wood Mackenzie) · DEWA Shams Dubai compliance guide (SurgePV) · DEWA eligible equipment list page (fetch failed) · DEWA DRRG 2017 equipment list PDF (not opened) · Ingeteam DEWA qualification · REFUsol DEWA listing · Schneider Electric DEWA approval · Solplanet DEWA approval (MESIA) · ECRA Regulatory Framework for Small-Scale Solar PV (ESMAP mirror PDF) · SASO IECEE recognition certification (SGS) · Enerwhere Buhasa camp hybrid · Enerwhere DIFC battery (pv-magazine) · Tanweer 200 MW hybrid plans (Oman Observer) · Oman 11-site 160 MW hybrid tender (pv-magazine) · Elum compatibility tool (rendered empty) · Elum llms.txt site index · Marizad–Elum exclusive agency (Arabic, Almal News)